

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

I B. SREENITHA (192312203) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

B. SREENITHA

Signature of the student:

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns a general rule or model from previously observed patient data and uses it to predict the outcome for new patients.

For this problem, a **supervised learning classification approach** can be used.

- **Target variable:** Diabetes — 1 = Diabetic, 0 = Non-diabetic
- **Important features:**
 1. **Glucose level** – High glucose is a strong indicator of diabetes.
 2. **BMI** – Higher BMI can be associated with increased diabetes risk.
 3. **Blood pressure** – Abnormal blood pressure can contribute to metabolic and cardiovascular risk.
 4. **Age** – Diabetes risk generally increases with age.
 5. **Family history** – A family history of diabetes indicates increased genetic risk.

The model learns the relationship between these features and the diabetes outcome and then predicts whether a new patient is diabetic.

(b) Handling Class Imbalance

If there are many more non-diabetic patients than diabetic patients, the dataset is **imbalanced**.

I would use **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE creates synthetic examples of the minority diabetic class instead of simply duplicating existing records.

Advantages:

- Increases the number of diabetic samples.
- Helps the classifier learn minority-class patterns.
- Reduces bias toward the majority class.
- Can improve **Recall and F1-score** for diabetic patients.

For medical diagnosis, recall is especially important because missing a diabetic patient (**False Negative**) can be clinically harmful.

(c) 70:30 Split and Preprocessing

The dataset can be divided into:

- **70% training data**
- **30% testing data**

A 70:30 split provides enough data for training while retaining a sufficiently large independent test set for evaluating the model.

For a medical dataset, I would use **stratified splitting** so that the diabetic/non-diabetic ratio remains approximately the same in both sets.

Preprocessing:

1. Handle missing values.
2. Remove duplicate or invalid records.
3. Apply **SMOTE only to the training data**.
4. Normalize numerical features if required.
5. Encode categorical variables such as family history.
6. Split into training and testing sets.

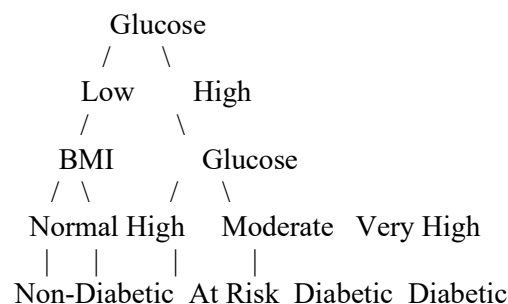
Normalization prevents features with larger numerical ranges from dominating models that are sensitive to feature scale, while encoding converts categorical information into numerical form.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree and First Three Levels

A Decision Tree recursively divides patients based on feature values.

A possible tree structure is:



The **root node** should be selected using the feature having the highest **Information Gain** or the lowest **Gini impurity** after splitting.

For example:
Information Gain:

Glucose = 0.35
BMI = 0.18
Age = 0.12
Blood Pressure = 0.09

Therefore, **Glucose** would be selected as the root because it provides the largest reduction in uncertainty.
The exact numerical Information Gain/Gini values must be calculated from the actual dataset. The values above are an illustration, not measured results.

(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

Precision

Recall

F1-score

In this medical application, **False Negatives are particularly important**.

A False Negative means:

The patient is actually diabetic, but the model predicts non-diabetic.

To reduce False Negatives:

- Increase the importance of the diabetic class using **class weights**.
- Tune the classification threshold.
- Use SMOTE on training data.
- Optimize for **Recall** rather than only accuracy.
- Validate the model using cross-validation.

Clinically, reducing False Negatives is important because an undiagnosed diabetic patient may not receive timely medical evaluation or treatment.

(c) Pre-Pruning vs Post-Pruning

Pre-pruning stops tree growth early using parameters such as:

- max_depth
- min_samples_split
- min_samples_leaf

Post-pruning first grows the tree and then removes branches that contribute little predictive value.

For Decision Trees, **cost-complexity pruning** can be represented as:

where:

- = prediction error
- = number of terminal nodes
- = complexity parameter

Model	Expected behavior
Pre-pruned Tree	Simpler and faster
Unpruned Tree	May overfit
Post-pruned Tree	Better generalization and interpretability

For clinical deployment, a **properly post-pruned tree** is generally preferable if validation shows comparable or better performance because it reduces unnecessary complexity and overfitting.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

I would use **Logistic Regression** because the target variable is binary:

- 0 = Non-diabetic
- 1 = Diabetic

The logistic function is:

where:

The model produces the probability that a patient has diabetes.

Comparison

Metric	Decision Tree	Logistic Regression
Accuracy	Calculate from dataset	Calculate from dataset
AUC-ROC	Calculate from dataset	Calculate from dataset
Interpretability	High	High

Non-linear relationships	Good	Limited
Probability-based risk	Moderate	Good

The exact Accuracy and AUC-ROC values should be obtained by running both models on the provided dataset. The assessment document does not provide those measured values.

CO4_AT1.docx

When Logistic Regression is preferable:

- When relationships are approximately linear.
- When probability/risk estimation is important.
- When model coefficients need to be interpreted.
- When a simpler statistical model is preferred.

(b) Feature Importance

For the **Decision Tree**, feature importance can be obtained from the reduction in impurity produced by each feature.

For **Logistic Regression**, feature importance can be examined using the magnitude of standardized coefficients.

A reasonable expected top-three set for this diabetes problem is:

Decision Tree:

1. Glucose
2. BMI
3. Age

Logistic Regression:

1. Glucose
2. BMI
3. Age

However, these rankings must be calculated from the actual dataset before reporting them as experimental results.

If both models identify similar predictors, it indicates that those features have a consistent relationship with diabetes risk.

Task 4: Reinforcement Learning for Treatment Recommendation

The assessment asks us to model treatment recommendation as an **MDP** and use Q-Learning for at least five patient episodes.

CO4_AT1.docx

(a) MDP Model

An MDP consists of:

States

Patient health states can be represented as:

S0 = Healthy / Low Risk

S1 = At Risk

S2 = High Risk

S3 = Diabetic / Poor Control

S4 = Improving

Actions

A0 = Diet

A1 = Exercise

A2 = Medication

A3 = Monitor

Reward Function

Example:

Health improves significantly → +10

Health improves slightly → +5

No significant change → 0

Health worsens → -10

Unsafe outcome → -20

The reward encourages the agent to select actions that lead to improved patient health.

Transition Dynamics

Example:

High Risk + Diet

↓

At Risk / Improving

High Risk + Exercise

↓
At Risk / Improving

Diabetic + Medication

↓
Improving / Controlled

Any state + Monitor

↓
State assessment / possible transition

In a real healthcare system, transition probabilities must come from validated clinical evidence and should **not** be invented by an AI system.

(b) Q-Learning

The Q-Learning equation is:

Where:

- = current Q-value
- = learning rate
- = reward
- = discount factor
- = next state

Assume:

Initial Q-table

State	Diet	Exercise	Medication	Monitor
Low Risk	0	0	0	0
At Risk	0	0	0	0
High Risk	0	0	0	0
Diabetic	0	0	0	0

Iteration 1

Suppose:

State = High Risk

Action = Diet

Reward = +5

Next State = At Risk

Since all initial Q-values are 0:

Another update:

State = At Risk

Action = Exercise

Reward = +5

Next State = Improving

Third update:

State = Diabetic

Action = Medication

Reward = +10

Next State = Improving

Iteration 2

Suppose:

High Risk → Diet → At Risk

Reward = +5

Now:

Similarly, if:

At Risk → Exercise → Improving

Reward = +5

then:

If:

Diabetic → Medication → Improving

Reward = +10

then:

Example Updated Q-table

State	Diet	Exercise	Medication	Monitor
Low Risk	2.0	1.5	0.5	3.0
At Risk	2.0	3.75	1.0	2.5
High Risk	3.875	2.0	3.0	2.5
Diabetic	1.0	2.0	5.0	2.0

These numbers are **illustrative Q-learning calculations**, not experimental results from an actual patient dataset.

Convergence Criterion

The algorithm can be considered converged when:

for example:

for several consecutive episodes.

The assessment specifically requires at least **5 patient episodes**, so the implementation should run five or more episodes and record the Q-table updates.

CO4_AT1.docx

(c) Final Learned Policy

The policy is obtained using:

From the illustrative Q-table:

Patient State	Best Action
Low Risk	Monitor
At Risk	Exercise
High Risk	Diet
Diabetic	Medication

Important: These are only example learned actions for demonstrating Q-Learning. They must **not** be treated as actual medical treatment recommendations.

Ethical Considerations

1. Patient Safety

The RL agent should never independently prescribe treatment. Medical professionals must remain responsible for final decisions.

2. Bias

If the training data contains demographic or socioeconomic bias, the RL agent may produce biased recommendations.

3. Explainability

The system should explain why a particular action was recommended.

4. Human Oversight

A doctor should review recommendations before implementation.

5. Privacy

Patient information must be securely stored and protected.

6. Validation

The RL agent must undergo extensive clinical validation before real-world deployment.

Final Conclusion

The proposed system combines **inductive supervised learning, Decision Trees, Logistic Regression, and Reinforcement Learning** to address diabetes diagnosis and treatment recommendation. Decision Trees provide interpretable diagnostic rules, while Logistic Regression provides probability-based risk estimation. Q-Learning can model sequential decision-making, but in a healthcare environment the agent must operate under strict clinical constraints, safety checks, privacy protection, bias monitoring, and human supervision.